

Phoebe Ly

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A forward-thinking spatial data analyst and product engineer, driven by the opportunity to transform complex datasets into practical, scalable, and meaningful innovations. Experienced in broad data science methodologies with a specialized focus on geospatial applications in domains such as agriculture, public health, aviation, and more. High achiever who excels in collaborative environments, contributing to team success and earning prestigious recognition, such as the Gavi 2024 INFUSE Pacesetter and the 100+ Startup of the Year Award.

EDUCATION

University of California, Riverside (UCR)

2017 - 2021

Bachelor of Science, Major in Statistics, Minor in Computer Science

Honors Thesis: Contribution of Plant Canopies to Soil Organic Carbon Along an Elevation Gradient

TECHNICAL SKILLS

Languages: Python, R, SQL (Proficient); C++, SAS, MATLAB, HTML, CSS, Javascript, Shell Scripting (Experience)

Software: RStudio, JupyterLab, QGIS, Tableau, Docker

Platforms: GitHub, MLFlow, Google Earth Engine, Google Cloud Platform, AWS

Specializations: GIS Analysis, Remote Sensing, Spatial Statistics, Data Visualization, Machine Learning, Deep Learning

Types of data: Satellite (Vectors/Rasters), GPS, Road/Network, Environmental, Inventory, Text

WORK EXPERIENCE

Geospatial Data Analyst - Atlas AI, Remote

2022 - present

- Collaborated with internal and client teams to define analytical needs and frame problems, acquiring relevant geospatial and sensor-based data from public and third-party sources.
- Utilized tools like GEE, Python, and QGIS to productize research outputs and refine Machine Learning models for new contexts.
- Developed and deployed a range of gradient boosting Machine Learning models, utilizing MLFlow for comprehensive version tracking from initiation to completion.
- Managed multiple projects at a time (self-directed and collaborative) while meeting tight deadlines.
- Secured an additional \$100,000 in project funding by delivering valuable results and proof of concept.
- Contributed to research proposals, customer solutions, and demos, while supporting technical sales and content creation for internal and external stakeholders.

Research Assistant / Student - Center of Food Security & Environment, Stanford University

2021 - 2022

- Investigated the impact of soil quality on children's health and nutrition in developing countries using remote sensing techniques.
- Conducted extensive literature reviews on remote sensing applications and micronutrient deficiencies to support research objectives.
- Sampled and extracted Sentinel-2 data with GEE and web-scraped data tables using Python's Selenium package.
- Developed R scripts to create exploratory maps and graphs, and implemented linear and random forest models with cross-validation for data analysis.
- Authored the scientific abstract and presented a poster at the 2021 American Geophysical Union Conference.

Student Researcher - Soil Biogeochemistry Lab, UC Riverside

2019 - 2021

- Analyzed X-ray Fluorescence soil data to find trends and characterize soil composition for research.
- Automated the generation of user-friendly graphs for audiences with limited statistical expertise.
- Collaborated effectively in an interdisciplinary team, gaining insights into an unfamiliar field.
- Assisted with multiple projects involving data cleaning and analysis.

Data Analyst Intern - Hyperion Development, Cape Town, South Africa

Summer 2019

- Automated an internal employee progress tracking tool to generate dashboards and report key metrics.

- Revised and reformatted the Software Engineering and Data Science Bootcamp courses to improve student learning outcomes.
- Explored various data visualization tools, including Metabase and Tableau, to identify cost-effective solutions for communicating analysis to employees.

LEADERSHIP EXPERIENCE

Women in Data Science (WiDS) Ambassador, Stanford University **2020 - 2022**

- Established the inaugural Riverside WiDS Regional event.
- Organized event logistics and addressed all guest speaker inquiries.
- Moderated panels and hosted "Intro to R" workshops for undergraduate students.

President, Highlander Statistics Society, UCR **2019 - 2021**

- Integrated, organized, and led coding workshops for students to learn R at club meetings.
- Held bi-weekly club meetings to support undergraduate students find opportunities in their careers and strengthen their technical skills.
- Coordinated with faculty members and alumni to visit tech companies and set up presentations.

Representative, Diversity, Equity, Inclusion Committee (DEI), UCR **2021**

- Brainstormed ideas and initiatives to build DEI efforts within the Statistics Department.
- Executed virtual events to bridge the gap between students and faculty.

HONORS & ASSOCIATIONS

Outstanding Student in Statistics Award, ASA OCLB Chapter **January 2022**

Honors Excellence In Research Scholarship Award, UCR **January 2021**

University Honors, UCR **August 2018 - June 2021**